Berliner Hochschule Für Technik

Department VII – Information and Communication Engineering

Expose For Master's Thesis

Integrated Edge-to-Cloud Systems: Federated Learning for Sensor Fusion and Object Recognition

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1. Introduction and Motivation:

Most current robotics systems depend primarily on cameras (vision) to understand the world. While this works well in ideal conditions, it creates vulnerabilities in low-light environments, fog, smoke, or physical obstructions where robots become effectively blind. This limitation reminds one of human perception — when we cannot see, we rely on hearing to understand our surroundings.

The motivation for this thesis is to give robots these same capabilities by adding microphones and using sensor fusion — combining sound and vision — to recognize objects more robustly. Processing sound and vision requires learning from raw data collected at the edge devices. However, there is a fundamental trade-off to investigate: how do we balance **privacy, power consumption, and model accuracy** when choosing where and how training happens?

This thesis does not assume one approach is best. Instead, it will compare three distinct strategies:

- **Centralized Learning:** Raw data sent to cloud; training happens on server.
- **Hybrid Learning (STFT):** Audio converted to spectrograms locally, then sent to cloud for training.
- **Federated Learning:** Local training on edge devices, only model weights shared.

Each approach has different implications for privacy, power consumption, bandwidth usage, model quality, and the ability to adapt to changing environments. The goal is to investigate these trade-offs systematically and then select one strategy for deeper implementation based on empirical findings.

2. Problem Statement:

Camera-only robotic systems lack robustness because they fail in low-light or visually obstructed environments. Adding audio sensors improves perception but raises new questions about where and how machine learning should happen.

The key challenge is: How can we implement robust multimodal sensor fusion while balancing privacy, power consumption, and model accuracy across different training strategies?

This is not a question with a predetermined answer. It requires comparing approaches and understanding their trade-offs in realistic conditions.

Objectives:

The objective of this thesis is to compare three different learning strategies — centralized, hybrid (STFT-based), and federated — across multiple evaluation dimensions, and then based on that comparison, select one strategy for practical implementation.

Specific objectives:

1. Develop a multimodal edge system

Integrate camera and microphone sensors into two Raspberry Pi edge devices placed in different environments, connected to a server for collaborative model training.

2. Implement three strategies for comparison

- Strategy A (Centralized): Raw audio and video sent to cloud for training.
- Strategy B (Hybrid STFT): Audio converted to spectrograms locally, then sent to cloud.
- Strategy C (Federated): Local training on each Pi, only model weights shared.

3. Design an efficient audio processing pipeline

Implement STFT-based spectrogram generation on the Raspberry Pi — lightweight and sufficient for environmental sound detection (cars, machinery). Implement audio-visual sensor fusion

Develop models that combine visual and auditory inputs for object recognition, comparing how each strategy affects fusion performance.

4. Evaluate and compare all three strategies

Measure accuracy, privacy, power consumption, bandwidth usage, convergence behavior, and model freshness over time.

5. Select one strategy for final implementation

Based on comparison results, choose the most suitable approach and implement it as the practical contribution of the thesis.

3. Literature Review:

3.1 Sensor Fusion (Vision and Auditory)

Traditional robotic perception relies heavily on vision. However, cameras fail in challenging environments such as low-light areas, smoke, or physical obstructions. Sensor fusion combines vision and sound to give robots human-like multi-sensory capabilities. Research suggests that "feature-level fusion" where data from both sensors is integrated within

a neural network — enables more robust object recognition. When visual input is compromised, the system can rely on auditory signals to maintain accuracy.

3.2 Federated Learning and Distributed Training

Federated Learning (FL) is a decentralized approach where models are trained locally on edge devices, and only weight updates are shared with a central server. This preserves privacy since raw data never leaves the device. However, FL introduces challenges:

Weight averaging and initialization: FedAvg works because all clients start with the same server-initialized model before each training round. If clients were initialized independently, weight averaging would be meaningless due to the permutation symmetry problem — networks can learn the same features with neurons in different positions, making direct averaging invalid. The shared initialization anchors the filters to the same feature space.

Non-IID data and client drift: Even with identical initialization, when client data distributions are different (non-IID), local training pulls models in different directions. This client drift can degrade FedAvg accuracy significantly — studies show drops of up to 55% under strongly non-IID conditions. These are open research questions worth empirical investigation.

3.3 Data Parallelism and Edge Computing

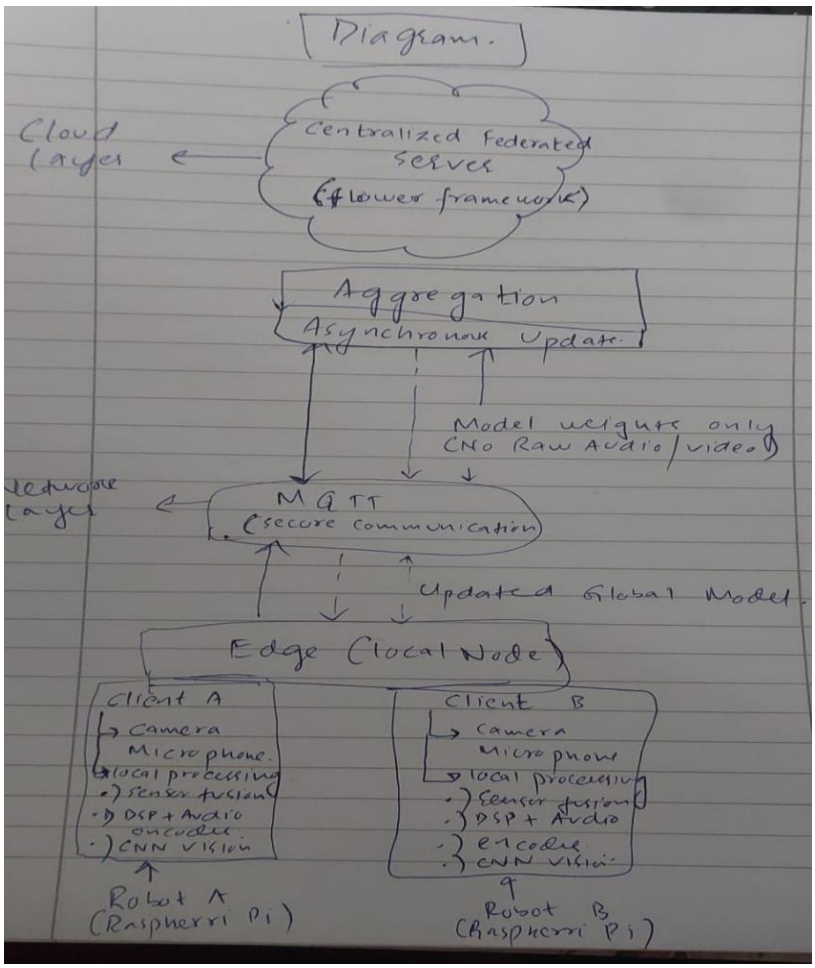
When dealing with multiple robotic agents, data parallelism enables distributed training. Each robot trains on its own local dataset, and their collective intelligence can be merged. Edge devices like Raspberry Pi have limited resources, so lightweight frameworks like TensorFlow Lite are essential. The choice between edge-based and cloud-based training has implications for power consumption, latency, and model freshness — how quickly the model adapts to environmental changes

3.4 Comparison: Centralized vs. Federated Learning:

Feature	Centralized Learning	Hybrid Learning (STFT)	Federated Learning
Data Privacy	Low — raw data leaves device	Medium — only spectrograms sent	High — raw data never leaves
Power Consumption	Low — Pi does inference only	Low — Pi does STFT + inference	High — Pi does training + inference
Bandwidth	High (event-based can reduce)	Medium — spectrograms only depend on resolution	Low — only models weight occasionally
Model Quality	Best — server sees all data	Good — server sees processed data	Uncertain — may degrade with non-IID
Model Freshness	Low — waits for cloud updates	Low — waits for cloud updates	High — continuous local adaptation
Implementation Complexity	Low	Medium	High

4.Proposed Methodology:

4.1 System Architecture



Workflow:

An architecture diagram has been (see Figure) showing two Raspberry Pi 4 edge nodes with camera and microphone sensors. The system supports three different data paths representing each strategy: Strategy A (Centralized): Raw audio and labels (obtained from vision) are transmitted to cloud server for training. Strategy B (Hybrid STFT): Audio converted to spectrograms locally, then sent to cloud. Strategy C (Federated): Local training on each Pi, only model weights shared via MQTT to the Flower framework for aggregation. The updated global model is then redistributed to the edge devices. MQTT ensures secure, lightweight communication between layers

. **4.2 Learning Framework (Hardware & Software Requirements)**

Hardware Requirements	**Software Requirements**
Edge Robot: 2× Raspberry Pi 4 (8 GB)	ML Framework: TensorFlow Lite
Camera Module 3 (current generation)	Communication: MQTT
I2S Digital Microphone (or lab analog mic interfaces if available)	DSP Libraries: Python, Librosa, NumPy (STFT only)
Network: 5 GHz Wi-Fi Router	FL Framework: Flower (flwr) open source, supports FedAvg
Cloud Server: AWS Free Tier (no cost)	Aggregation: FedAvg algorithm

Optional: Battery power banks for realistic power measurement	Visualization: Matplotlib for spectrograms

4.3 Label Acquisition Strategy

To avoid manual labeling (thousands of samples would take weeks), we use a pre-trained vision model (YOLOv8) running locally on each Raspberry Pi. It provides real-time object detections from the camera feed, which serve as automatic labels for corresponding audio segments. This enables two scenarios suggested (i) updating both audio and vision models collaboratively, and (ii) keeping vision fixed while updating only audio. Both scenarios will be investigated across all three strategies (centralized, hybrid STFT, federated), keeping the focus on comparing the learning approaches themselves.

5. Evaluation Metrics

Metric	Description
Accuracy	Object recognition accuracy on a held-out test set, comparing performance across all three strategies.
Privacy	Assessment of what data leaves the device: raw audio (Strategy A), spectrograms (Strategy B), or model weights only (Strategy C).
Power Consumption	Measured in watts. For Strategy C, this will be measured separately for inference-only vs. inference+training modes to account for the additional load of local training.
Bandwidth	MB transmitted per training cycle for each strategy, comparing raw data vs. spectrograms vs. model weights.
Latency	Inference time on Raspberry Pi (ms). This will be measured consistently across all strategies since inference happens on the edge in all cases.
Memory Usage	RAM usage on Raspberry Pi. Measured separately for: (a) inference-only mode (all strategies), and (b) inference+training mode (Strategy C only), as memory consumption varies significantly between these scenarios.
Model Freshness	How quickly the model adapts to environmental changes — simulated by introducing new sound patterns mid-experiment and measuring detection accuracy over time.
Convergence (Strategy C only)	For federated learning, measuring client drift and FedAvg performance under non-IID conditions.

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